Important Terms Provide definitions or explain the meaning of the following terms:

admittance Y
air line
attenuation constant α
bounce diagram
characteristic impedance Z_0
coaxial line
complex propagation constant γ
conductance G
current maxima and minima
dispersive transmission line
distortionless line
effective relative permittivity ε_{eff}
guide wavelength λ
higher-order transmission lines
impedance matching
in-phase
input impedance Z_{in}
load impedance Z_{L}
lossless line
lumped-element model
matched transmission line
matching network
microstrip line
normalized impedance
normalized load reactance x_{L}
normalized load resistance r_{L}
open-circuited line
optical fiber
parallel-plate line
perfect conductor
perfect dielectric
phase constant β
phase opposition
phase-shifted reflection coefficient Γ_{d}
quarter-wave transformer
short-circuited line
single-stub matching
slotted line
Smith chart
standing wave
standing-wave pattern
surface resistance R_{s}
susceptance B
SWR circle
telegrapher's equations
TEM transmission lines
time-average power P_{av}
transient response
transmission-line parameters
two-wire line
unit circle
voltage maxima and minima
voltage reflection coefficient Γ
voltage standing-wave ratio (VSWR or SWR) S
wave equations
wave impedance $Z(d)$
waveguide
WTG and WTL

PROBLEMS

Sections 2-1 to 2-4: Transmission-Line Model

2.1 A transmission line of length l connects a load to a sinusoidal voltage source with an oscillation frequency f. Assuming that the velocity of wave propagation on the line is c, for which of the following situations is it reasonable to ignore the presence of the transmission line in the solution of the circuit?

*(a) $l = 20$ cm, $f = 20$ kHz
(b) $l = 50$ km, $f = 60$ Hz
*(c) $l = 20$ cm, $f = 600$ MHz
(d) $l = 1$ mm, $f = 100$ GHz

2.2 A two-wire copper transmission line is embedded in a dielectric material with $\varepsilon_{\text{r}} = 2.6$ and $\sigma = 2 \times 10^{-6}$ S/m. Its wires are separated by 3 cm and their radii are 1 mm each.

(a) Calculate the line parameters R', L', G', and C' at 2 GHz.
(b) Compare your results with those based on Module 2.1. Include a printout of the screen display.

2.3 Show that the transmission-line model shown in **Fig. P2.3** yields the same telegrapher's equations given by Eqs. (2.14) and (2.16).

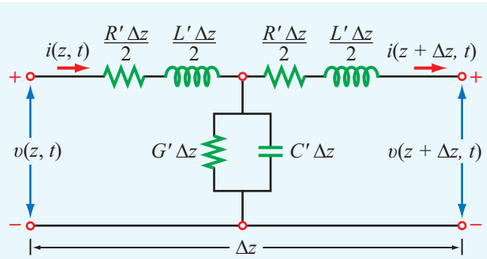


Figure P2.3 Transmission-line model for Problem 2.3.

*Answer(s) available in Appendix E.

*2.4 A 1 GHz parallel-plate transmission line consists of 1.2-cm wide copper strips separated by a 0.15-cm thick layer of polystyrene. Appendix B gives $\mu_{\text{c}} = \mu_0 = 4\pi \times 10^{-7}$ (H/m) and $\sigma_{\text{c}} = 5.8 \times 10^7$ (S/m) for copper, and $\varepsilon_{\text{r}} = 2.6$ for polystyrene. Use **Table 2-1** to determine the line parameters of the transmission line. Assume that $\mu = \mu_0$ and $\sigma \approx 0$ for polystyrene.

2.5 For the parallel-plate transmission line of Problem 2.4, the line parameters are given by $R' = 1\ \Omega$/m, $L' = 167$ nH/m, $G' = 0$, and $C' = 172$ pF/m. Find α, β, u_{p}, and Z_0 at 1 GHz.

重要术语 提供以下术语的定义或解释其含义：

导纳 Y 空气线路衰减常数 α 弹跳图 特性阻抗 Z_0 同轴线复传播常数 γ 电导 G 电流最大值和最小值色散传输线 无失真线路 有效相对介电常数 ε_{eff} 波导波长 λ 高阶传输线阻抗匹配 同相输入阻抗 Z_{in} 负载阻抗 Z_{L} 无损耗线路集总元件模型 匹配传输线 匹配网络 微带线 归一化阻抗 归一化负载电抗 x_{L} 归一化负载电阻 r_{L} 开路线 光纤 平行板线 理想导体 理想介电相位常数 β 反相移相反射系数 Γ_{d} 四分之一波变压器 短路线 单截线匹配 开槽线 史密斯圆图 驻波 驻波图 表面电阻 R_{s} 电纳 B SWR 圆电报方程 TEM 传输线 时间平均功率 P_{av} 瞬态响应 传输线参数 两线传输线单位圆 电压最大值和最小值 电压反射系数 Γ 电压驻波比（VSWR 或 SWR） S 波方程 波阻抗 Z(d) 波导 WTG 和 WTL

问题

第 2-1 节到 2-4 节：传输线模型

2.1 长度为 l 的传输线将负载连接到振荡频率为 f 的正弦电压源。假设波在线路上的传播速度为 c，对于以下哪种情况，在电路解中忽略传输线的存在是合理的？

*(a) l = 20 厘米，f = 20 kHz
(b) l = 50 公里，f = 60 Hz
*(c) l = 20 厘米，f = 600 MHz
(d) l = 1 毫米，f = 100 GHz

2.2 将两线铜传输线嵌入 $\varepsilon_{\text{r}} = 2.6$ 和 $\sigma = 2 \times 10^{-6}$ S/m 的介电材料中。其电线间隔 3 厘米，半径各 1 毫米。

(a) 计算 2 GHz 下的线路参数 R′、L′、G′ 和 C′。 (b) 将您的结果与基于模块 2.1 的结果进行比较。包括屏幕显示的打印输出。

2.3 表明图 P2.3 中所示的传输线模型产生与方程 1 给出的相同的电报员方程。 （2.14）和（2.16）。

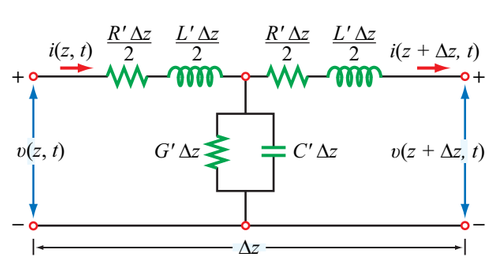


图 P2.3 问题 2.3 的传输线模型。

*Answer(s) available in Appendix E.

*2.4 1 GHz 平行板传输线由 1.2 厘米宽的铜带组成，铜带之间被 0.15 厘米厚的聚苯乙烯层隔开。附录 B 给出了铜的 $\mu_{\text{c}} = \mu_0 = 4\pi \times 10^{-7}$ (H/m) 和 $\sigma_{\text{c}} = 5.8 \times 10^7$ (S/m)，以及聚苯乙烯的 $\varepsilon_{\text{r}} = 2.6$。使用表2-1确定传输线的线路参数。假设聚苯乙烯的 $\mu = \mu_0$ 和 $\sigma \approx$ 为 0。

2.5 对于问题 2.4 的平行板传输线，线参数由 $R' = 1\ \Omega$/m、$L' = 167$ nH/m、$G' = 0$ 和 $C' = 172$ pF/m 给出。查找 1 GHz 处的 α、β、u_{p} 和 Z_0。

2.6 A coaxial line with inner and outer conductor diameters of 0.5 cm and 1 cm, respectively, is filled with an insulating material with $\varepsilon_r = 4.5$ and $\sigma = 10^{-3}$ S/m. The conductors are made of copper.

(a) Calculate the line parameters at 1 GHz.

(b) Compare your results with those based on Module 2.2. Include a printout of the screen display.

2.7 Find α, β, u_p, and Z_0 for the coaxial line of Problem 2.6. Verify your results by applying Module 2.2. Include a printout of the screen display.

*2.8 Find α, β, u_p, and Z_0 for the two-wire line of Problem 2.2. Compare results with those based on Module 2.1. Include a printout of the screen display.

Section 2-5: Microstrip

2.9 A lossless microstrip line uses a 1-mm wide conducting strip over a 1-cm thick substrate with $\varepsilon_r = 2.5$. Determine the line parameters, ε_{eff}, Z_0, and β at 10 GHz. Compare your results with those obtained by using Module 2.3. Include a printout of the screen display.

*2.10 Use Module 2.3 to design a 100 Ω microstrip transmission line. The substrate thickness is 1.8 mm and its $\varepsilon_r = 2.3$. Select the strip width w, and determine the guide wavelength λ at $f = 5$ GHz. Include a printout of the screen display.

2.11 A 50 Ω microstrip line uses a 0.6 mm alumina substrate with $\varepsilon_r = 9$. Use Module 2.3 to determine the required strip width w. Include a printout of the screen display.

2.12 Generate a plot of Z_0 as a function of strip width w over the range from 0.05 mm to 5 mm for a microstrip line fabricated on a 0.7-mm thick substrate with $\varepsilon_r = 9.8$.

Section 2-6: Lossless Line

2.13 In addition to not dissipating power, a lossless line has two important features: (1) it is dispersionless (u_p is independent of frequency), and (2) its characteristic impedance Z_0 is purely real. Sometimes, it is not possible to design a transmission line such that $R' \ll \omega L'$ and $G' \ll \omega C'$, but it is possible to choose the dimensions of the line and its material properties to satisfy the condition

$$R'C' = L'G' \qquad \text{(distortionless line)}$$

Such a line is called a ***distortionless*** line, because despite the fact that it is not lossless, it nonetheless possesses the previously mentioned features of the lossless line. Show that for a distortionless line,

$$\alpha = R'\sqrt{\frac{C'}{L'}} = \sqrt{R'G'},$$
$$\beta = \omega\sqrt{L'C'},$$
$$Z_0 = \sqrt{\frac{L'}{C'}}.$$

*2.14 For a distortionless line (see Problem 2.13) with $Z_0 = 50\ \Omega$, $\alpha = 20$ (mNp/m), and $u_p = 2.5 \times 10^8$ (m/s), find the line parameters and λ at 100 MHz.

2.15 Find α and Z_0 of a distortionless line whose $R' = 2\ \Omega$/m and $G' = 2 \times 10^{-4}$ S/m.

*2.16 A transmission line operating at 125 MHz has $Z_0 = 40\ \Omega$, $\alpha = 0.02$ (Np/m), and $\beta = 0.75$ rad/m. Find the line parameters R', L', G', and C'.

2.17 Using a slotted line, the voltage on a lossless transmission line was found to have a maximum magnitude of 1.5 V and a minimum magnitude of 0.6 V. Find the magnitude of the load's reflection coefficient.

*2.18 Polyethylene with $\varepsilon_r = 2.25$ is used as the insulating material in a lossless coaxial line with a characteristic impedance of 50 Ω. The radius of the inner conductor is 1.2 mm.

(a) What is the radius of the outer conductor?

(b) What is the phase velocity of the line?

2.19 A 50 Ω lossless transmission line is terminated in a load with impedance $Z_L = (30 - j50)\ \Omega$. The wavelength is 8 cm. Determine:

(a) The reflection coefficient at the load.

(b) The standing-wave ratio on the line.

(c) The position of the voltage maximum nearest the load.

(d) The position of the current maximum nearest the load.

(e) Verify quantities in parts (a)–(d) using Module 2.4. Include a printout of the screen display.

2.20 A 300 Ω lossless air transmission line is connected to a complex load composed of a resistor in series with an inductor, as shown in **Fig. P2.20**. At 5 MHz, determine: (a) Γ, (b) S, (c) location of voltage maximum nearest to the load, and (d) location of current maximum nearest to the load.

2.6 内、外导体直径分别为 0.5 cm 和 1 cm 的同轴线，填充有 $\varepsilon_r = 4.5$ 和 $\sigma = 10^{-3}$ S/m 的绝缘材料。导体由铜制成。

(a) 计算 1 GHz 下的线路参数。
(b) 将您的结果与基于模块 2.2 的结果进行比较。包括屏幕显示的打印输出。

2.7 求问题2.6的同轴线的α、β、u_p和Z_0。通过应用模块 2.2 验证您的结果。包括屏幕显示的打印输出。

*2.8 求问题 2.2 的两线线路的 α、β、u_p 和 Z_0。将结果与基于模块 2.1 的结果进行比较。包括屏幕显示的打印输出。

第 2-5 节：微带线

2.9 无损微带线在 1 厘米厚的基板上使用 1 毫米宽的导电带，$\varepsilon_r = 2.5$。确定 10 GHz 下的线路参数 ε_{eff}、Z_0 和 β。将您的结果与使用模块 2.3 获得的结果进行比较。包括屏幕显示的打印输出。

*2.10 使用模块 2.3 设计 100 Ω 微带传输线。基板厚度为1.8mm，其ε_r =2.3。选择条带宽度 w，并确定 f = 5 GHz 处的波导波长 λ。包括屏幕显示的打印输出。

2.11 50 Ω 微带线使用带有 $\varepsilon_r = 9$ 的 0.6 mm 氧化铝基板。使用模块 2.3 确定所需的带宽度 w。包括屏幕显示的打印输出。

2.12 对于在 0.7 毫米厚的基板上制作的微带线，ε_r = 为 9.8，生成 Z_0 作为带宽 w 的函数在 0.05 毫米到 5 毫米范围内的图。

第 2-6 节：无损线路

2.13 除了不耗散功率之外，无损线路还有两个重要特征：(1) 无色散（u_p 与频率无关），(2) 其特征阻抗 Z_0 是纯实数。有时，不可能设计出满足 $R' \ll \omega L'$ 和 $G' \ll \omega C'$ 的传输线，但可以选择线的尺寸及其材料属性来满足条件

$$R'C' = L'G' \qquad \text{(distortionless line)}$$

这样的线被称为无畸变线，因为尽管它不是无损的，但它仍然具有

前面提到了无损线路的特点。证明对于无畸变线，

$$\alpha = R'\sqrt{\frac{C'}{L'}} = \sqrt{R'G'},$$
$$\beta = \omega\sqrt{L'C'},$$
$$Z_0 = \sqrt{\frac{L'}{C'}}.$$

*2.14 对于 $Z_0 = 50\ \Omega$、$\alpha = 20$ (mNp/m) 和 $u_p = 2.5 \times 10^8$ (m/s) 的无失真线路（参见问题 2.13），求 100 MHz 处的线路参数和 λ。

2.15 求一条无畸变线的 α 和 Z_0，其 $R' = 2\ \Omega$/m 和 $G' = 2 \times 10^{-4}$ S/m。

*2.16 工作在 125 MHz 的传输线的 $Z_0 = 40\ \Omega$、$\alpha = 0.02$ (Np/m) 和 $\beta = 0.75$ rad/m。找到线参数 R′、L′、G′ 和 C′。

2.17 使用开槽线，发现无损传输线上的电压的最大值为 1.5 V，最小值为 0.6 V。求负载反射系数的大小。

*2.18 $\varepsilon_r = 2.25$ 的聚乙烯用作特性阻抗为 50 Ω 的无损同轴线中的绝缘材料。内导体的半径为1.2毫米。

(a) 外导体的半径是多少？ (b) 线的相速度是多少？

2.19 50 Ω 无损传输线的端接阻抗为 $Z_L = (30 - j50)\ \Omega$ 的负载。波长为8厘米。决定：

(a) 负载处的反射系数。
(b) 线上的驻波比。
(c) 最接近负载的电压最大值位置。
(d) 最接近负载的电流最大值的位置。 (e) 使用模块 2.4 验证 (a)–(d) 部分的数量。包括屏幕显示的打印输出。

2.20 将一条 300 Ω 无损空气传输线连接到由电阻与电感串联组成的复杂负载上，如图 P2.20 所示。在 5 MHz 处，确定：(a) Γ、(b) S、(c) 最接近负载的电压最大值位置，以及 (d) 最接近负载的电流最大值位置。

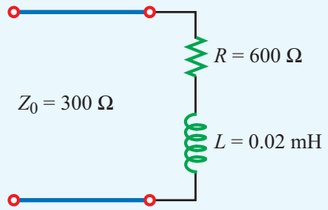


Figure P2.20 Circuit for Problem 2.20.

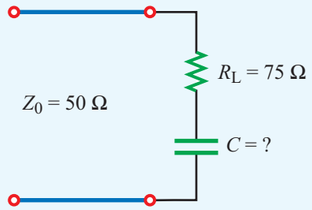


Figure P2.26 Circuit for Problem 2.26.

*2.21 On a 150 Ω lossless transmission line, the following observations were noted: distance of first voltage minimum from the load = 3 cm; distance of first voltage maximum from the load = 9 cm; and $S = 3$. Find Z_L.

2.22 Using a slotted line, the following results were obtained: distance of first minimum from the load = 4 cm; distance of second minimum from the load = 14 cm; and voltage standing-wave ratio = 1.5. If the line is lossless and $Z_0 = 50\ \Omega$, find the load impedance.

*2.23 A load with impedance $Z_L = (25 - j50)\ \Omega$ is to be connected to a lossless transmission line with the characteristic impedance Z_0. Z_0 is chosen such that the standing-wave ratio is the smallest possible. What should Z_0 be?

2.24 A 50 Ω lossless line terminated in a purely resistive load has a voltage standing-wave ratio of 3. Find all possible values of Z_L.

2.25 Apply Module 2.4 to generate plots of the voltage standing-wave pattern for a 50 Ω line terminated in a load impedance $Z_L = (100 - j50)\ \Omega$. Set $V_g = 1$ V, $Z_g = 50\ \Omega$, $\varepsilon_r = 2.25$, $l = 40$ cm, and $f = 1$ GHz. Also determine S, d_{max}, and d_{min}.

2.26 A 50 Ω lossless transmission line is connected to a load composed of a 75 Ω resistor in series with a capacitor of unknown capacitance (Fig. P2.26). If at 10 MHz the voltage standing-wave ratio on the line was measured to be 3, determine the capacitance C.

Section 2-7: Wave and Input Impedance

*2.27 At an operating frequency of 300 MHz, a lossless 50 Ω air-spaced transmission line 2.5 m in length is terminated with an impedance $Z_L = (40 + j20)\ \Omega$. Find the input impedance.

2.28 A lossless transmission line of electrical length $l = 0.35\lambda$ is terminated in a load impedance as shown in Fig. P2.28. Find Γ, S, and Z_{in}. Verify your results using Modules 2.4 or 2.5. Include a printout of the screen's output display.

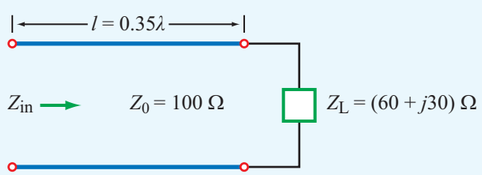


Figure P2.28 Circuit for Problem 2.28.

2.29 Show that the input impedance of a quarter-wavelength lossless line terminated in a short circuit appears as an open circuit.

2.30 Show that, at the position where the magnitude of the voltage on the line is a maximum, the input impedance is purely real.

2.31 A voltage generator with

$$v_g(t) = 5\cos(2\pi \times 10^9 t)\ \text{V}$$

and internal impedance $Z_g = 50\ \Omega$ is connected to a 50 Ω lossless air-spaced transmission line. The line length is 5 cm and the line is terminated in a load with impedance $Z_L = (100 - j100)\ \Omega$. Determine:

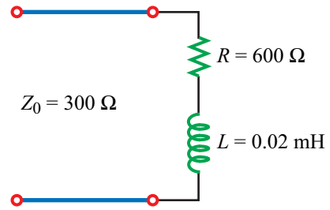


图 P2.20 问题 2.20 的电路。

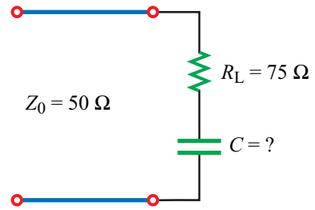


图 P2.26 问题 2.26 的电路。

*2.21 在 150 Ω 无损传输线上，注意到以下观察结果： 第一电压最小值与负载 = 的距离为 3 cm；第一个电压最大值距负载 = 的距离 9 cm；和 S = 3. 找到 Z_L。

2.22 使用开槽线，获得以下结果： 距负载 = 的第一个最小值的距离为 4 厘米；第二个最小值与负载的距离= 14 cm；电压驻波比= 1.5。如果线路是无损的并且 $Z_0 = 50\ \Omega$，则找到负载阻抗。

*2.23 阻抗为 $Z_L = (25 - j50)\ \Omega$ 的负载应连接到特性阻抗为 Z_0 的无损传输线上。选择 Z_0 以使驻波比尽可能最小。 Z_0 应该是什么？

2.24 端接于纯电阻负载的 50 Ω 无损线路的电压驻波比为 3。找出 Z_L 的所有可能值。

2.25 应用模块 2.4 生成终止于负载阻抗 $Z_L = (100 - j50)\ \Omega$ 的 50 Ω 线路的电压驻波模式图。设置 $V_g = 1$ V、$Z_g = 50\ \Omega$、$\varepsilon_r = 2.25$、$l = 40$ cm 和 f = 1 GHz。还要确定 S、d_{max} 和 d_{min}。

2.26 一条 50 Ω 无损传输线连接到一个由 75 Ω 电阻器与未知电容的电容器串联组成的负载（图 P2.26）。如果在 10 MHz 时测得线路上的电压驻波比为 3，则确定电容 C。

第 2-7 节：波形和输入阻抗

*2.27 在 300 MHz 的工作频率下，长度为 2.5 m 的无损 50 Ω 空气间隔传输线的端接阻抗为 $Z_L = (40 + j20)\ \Omega$。求输入阻抗。

2.28 电长度 $l = 0.35\lambda$ 的无损传输线的负载阻抗如图 P2.28 所示。找到 Γ、S 和 Z_{in}。使用模块 2.4 或 2.5 验证您的结果。包括屏幕输出显示的打印输出。

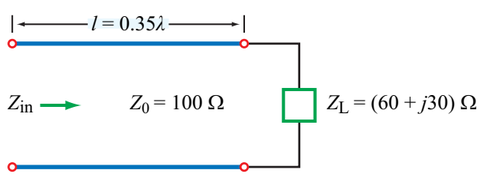


图 P2.28 问题 2.28 的电路。

2.29 证明以短路方式端接的四分之一波长无损线路的输入阻抗表现为开路。

2.30 证明，在线路上电压幅值最大的位置处，输入阻抗是纯实数。

2.31 电压发生器

$$v_g(t) = 5\cos(2\pi \times 10^9 t)\ \text{V}$$

内部阻抗Z_g =50Ω连接至50Ω无损空气间隔传输线。线路长度为 5 cm，线路终止于阻抗为 $Z_L = (100 - j100)\ \Omega$ 的负载。决定：

*(a) Γ at the load.

(b) Z_{in} at the input to the transmission line.

(c) The input voltage $\tilde{V}_i$ and input current $\tilde{I}_i$.

(d) The quantities in parts (a)–(c) using Modules 2.4 or 2.5.

2.32 A 6 m section of 150 Ω lossless line is driven by a source with

$$v_g(t) = 5\cos(8\pi \times 10^7 t - 30^\circ) \quad \text{(V)}$$

and $Z_g = 150\ \Omega$. If the line, which has a relative permittivity $\varepsilon_r = 2.25$, is terminated in a load $Z_L = (150 - j50)\ \Omega$, determine:

(a) λ on the line,

*(b) the reflection coefficient at the load,

(c) the input impedance,

(d) the input voltage $\tilde{V}_i$,

(e) the time-domain input voltage $v_i(t)$, and

(f) quantities in parts (a)–(d) using Modules 2.4 or 2.5.

2.33 Two half-wave dipole antennas, each with an impedance of 75 Ω, are connected in parallel through a pair of transmission lines, and the combination is connected to a feed transmission line, as shown in **Fig. P2.33**.

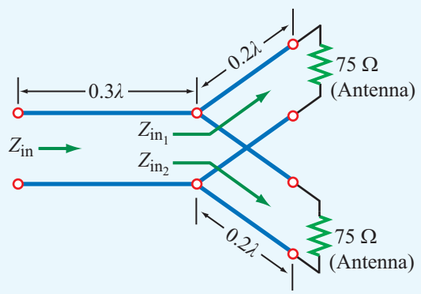


Figure P2.33 Circuit for Problem 2.33.

All lines are 50 Ω and lossless.

*(a) Calculate Z_{in_1}, the input impedance of the antenna-terminated line, at the parallel juncture.

(b) Combine Z_{in_1} and Z_{in_2} in parallel to obtain Z'_L, the effective load impedance of the feedline.

(c) Calculate Z_{in} of the feedline.

2.34 A 50 Ω lossless line is terminated in a load impedance $Z_L = (30 - j20)\ \Omega$.

(a) Calculate Γ and S.

(b) It has been proposed that by placing an appropriately selected resistor across the line at a distance d_{max} from the load (as shown in **Fig. P2.34(b)**), where d_{max} is the distance from the load of a voltage maximum, then it is possible to render $Z_i = Z_0$, thereby eliminating reflection back to the end. Show that the proposed approach is valid and find the value of the shunt resistance.

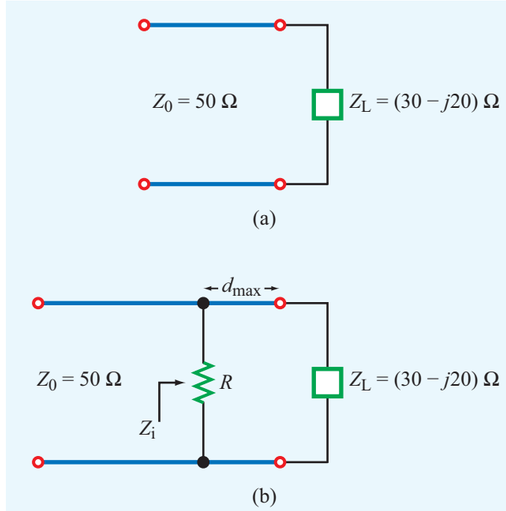


Figure P2.34 Circuit for Problem 2.34.

*2.35 For the lossless transmission line circuit shown in **Fig. P2.35**, determine the equivalent series lumped-element circuit at 400 MHz at the input to the line. The line has a characteristic impedance of 50 Ω and the insulating layer has $\varepsilon_r = 2.25$.

*(一个)在负载处。

(b)Z_{in} 位于传输线的输入端。

(c) 输入电压 $\tilde{V}_i$ 和输入电流 $\tilde{I}_i$。

(d) 使用模块 2.4 或 2.5 计算 (a)-(c) 部分的数量。

2.32 一段 6 m 的 150 Ω 无损线路由具有

$$v_g(t) = 5\cos(8\pi \times 10^7 t - 30^\circ) \quad \text{(V)}$$

和 $Z_g = 150\ \Omega$。如果相对介电常数 $\varepsilon_r = 2.25$ 的线路终止于负载 $Z_L = (150 - j50)\ \Omega$，则确定：

(a) λ 上线，

*(乙)负载下的反射系数，

(c) 输入阻抗，

(d) 输入电压 $\tilde{V}_i$，

(e) 时域输入电压 $v_i(t)$，以及 (f) 使用模块 2.4 或 2.5 计算 (a)–(d) 部分中的量。

2.33 两个阻抗为75Ω的半波偶极子天线通过一对传输线并联，并连接到馈电传输线，如图P2.33所示。

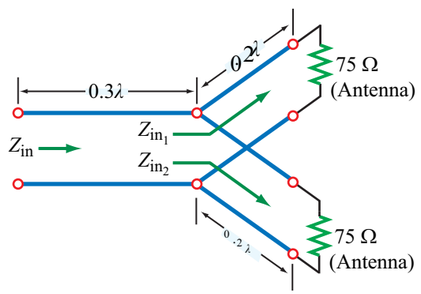


图 P2.33 问题 2.33 的电路。

所有线条均为 50 Ω 且无损。

*(a) 计算 Z_{in_1}，即天线端接线在平行接点处的输入阻抗。

(b) 并联组合 Z_{in_1} 和 Z_{in_2} 以获得 Z'_L，即馈线的有效负载阻抗。 (c) 计算馈线的 Z_{in}。

2.34 50 Ω 无损线路的端接负载阻抗为 $Z_L = (30 - j20)\ \Omega$。

(a) 计算Γ和S。

(b) 有人建议，通过在距负载 d_{max} 的距离处在线上放置一个适当选择的电阻器（如图 P2.34(b) 所示），其中 d_{max} 是距电压最大值负载的距离，则可以呈现 $Z_i = Z_0$，从而消除回到末端的反射。证明所提出的方法是有效的并找出分流电阻的值。

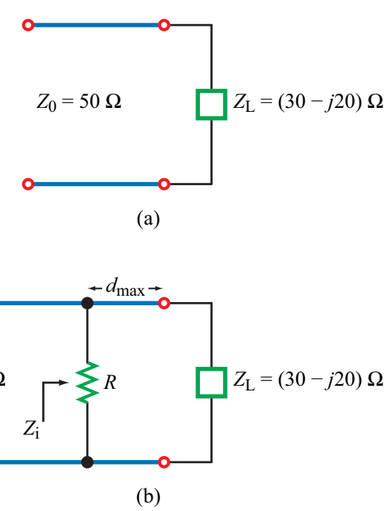


图 P2.34 问题 2.34 的电路。

*2.35 对于图 P2.35 所示的无损传输线电路，确定线路输入端 400 MHz 时的等效串联集总元件电路。线路的特性阻抗为50Ω，绝缘层的特性阻抗为ε_r =2.25。

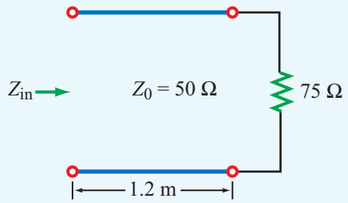


Figure P2.35 Circuit for Problem 2.35.

Section 2-8: Special Cases of the Lossless Line

2.36 At an operating frequency of 300 MHz, it is desired to use a section of a lossless 50 Ω transmission line terminated in a short circuit to construct an equivalent load with reactance $X = 40\ \Omega$. If the phase velocity of the line is $0.75c$, what is the shortest possible line length that would exhibit the desired reactance at its input? Verify your result using Module 2.5.

*2.37 A lossless transmission line is terminated in a short circuit. How long (in wavelengths) should the line be for it to appear as an open circuit at its input terminals?

2.38 The input impedance of a 31-cm long lossless transmission line of unknown characteristic impedance was measured at 1 MHz. With the line terminated in a short circuit, the measurement yielded an input impedance equivalent to an inductor with inductance of 0.064 μH, and when the line was open-circuited, the measurement yielded an input impedance equivalent to a capacitor with capacitance of 40 pF. Find Z_0 of the line, the phase velocity, and the relative permittivity of the insulating material.

*2.39 A 75 Ω resistive load is preceded by a $\lambda/4$ section of a 50 Ω lossless line, which itself is preceded by another $\lambda/4$ section of a 100 Ω line. What is the input impedance? Compare your result with that obtained through two successive applications of Module 2.5.

2.40 A 100 MHz FM broadcast station uses a 300 Ω transmission line between the transmitter and a tower-mounted half-wave dipole antenna. The antenna impedance is 73 Ω. You are asked to design a quarter-wave transformer to match the antenna to the line.

(a) Determine the electrical length and characteristic impedance of the quarter-wave section.

(b) If the quarter-wave section is a two-wire line with $D = 2.5$ cm and the wires are embedded in polystyrene with $\varepsilon_r = 2.6$, determine the physical length of the quarter-wave section and the radius of the two wire conductors.

2.41 A 50 Ω lossless line of length $l = 0.375\lambda$ connects a 300 MHz generator with $\widetilde{V}_g = 300$ V and $Z_g = 50\ \Omega$ to a load Z_L. Determine the time-domain current through the load for:

(a) $Z_L = (50 - j50)\ \Omega$

*(b) $Z_L = 50\ \Omega$

(c) $Z_L = 0$ (short circuit)

For part (a), verify your results by deducing the information you need from the output products generated by Module 2.4.

Section 2-9: Power Flow on Lossless Line

2.42 A generator with $\widetilde{V}_g = 300$ V and $Z_g = 50\ \Omega$ is connected to a load $Z_L = 75\ \Omega$ through a 50 Ω lossless line of length $l = 0.15\lambda$.

*(a) Compute Z_{in}, the input impedance of the line at the generator end.

(b) Compute $\widetilde{I}_i$ and $\widetilde{V}_i$.

(c) Compute the time-average power delivered to the line, $P_{in} = \frac{1}{2}\mathfrak{Re}[\widetilde{V}_i \widetilde{I}_i^*]$.

(d) Compute $\widetilde{V}_L$, $\widetilde{I}_L$, and the time-average power delivered to the load, $P_L = \frac{1}{2}\mathfrak{Re}[\widetilde{V}_L \widetilde{I}_L^*]$. How does P_{in} compare with P_L? Explain.

(e) Compute the time-average power delivered by the generator, P_g, and the time-average power dissipated in Z_g. Is conservation of power satisfied?

2.43 If the two-antenna configuration shown in **Fig. P2.43** is connected to a generator with $\widetilde{V}_g = 250$ V and $Z_g = 50\ \Omega$, how much average power is delivered to each antenna?

*2.44 For the circuit shown in **Fig. P2.44**, calculate the average incident power, the average reflected power, and the average power transmitted into the infinite 100 Ω line. The $\lambda/2$ line is lossless and the infinitely long line is slightly lossy. (*Hint:* The input impedance of an infinitely long line is equal to its characteristic impedance so long as $\alpha \neq 0$.)

2.45 The circuit shown in **Fig. P2.45** consists of a 100 Ω lossless transmission line terminated in a load with $Z_L = (50 + j100)\ \Omega$. If the peak value of the load voltage was measured to be $|\widetilde{V}_L| = 12$ V, determine:

*(a) the time-average power dissipated in the load,

(b) the time-average power incident on the line, and

(c) the time-average power reflected by the load.

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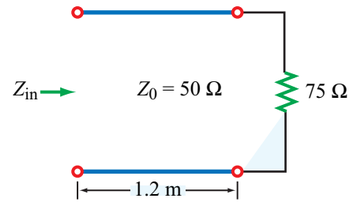


图 P2.35 问题 2.35 的电路。

第 2-8 节：无损线路的特殊情况

2.36 在 300 MHz 的工作频率下，希望使用一段以短路方式端接的无损 50 Ω 传输线来构建电抗 X = 40 Ω 的等效负载。如果线路的相速度为 0.75c，那么在其输入端表现出所需电抗的最短可能线路长度是多少？使用模块 2.5 验证您的结果。

*2.37 无损传输线以短路终止。线路应该有多长（以波长为单位）才能在其输入端子处显示为开路？

2.38 在 1 MHz 下测量了 31 厘米长、未知特性阻抗的无损传输线的输入阻抗。当线路端接短路时，测量产生的输入阻抗相当于电感为 0.064 μ H 的电感器，当线路开路时，测量产生的输入阻抗相当于电容为 40 pF 的电容器。求线的 Z_0、相速度和绝缘材料的相对介电常数。

*2.39 75 Ω 电阻负载之前是 50 Ω 无损线路的 λ /4 部分，而该线路本身又是 100 Ω 线路的另一个 λ /4 部分。输入阻抗是多少？将您的结果与通过两次连续应用模块 2.5 获得的结果进行比较。

2.40 100 MHz FM 广播电台在发射机和塔上安装的半波偶极天线之间使用 300 Ω 传输线。天线阻抗为73Ω。您需要设计一个四分之一波长变压器以使天线与线路相匹配。

(a) 确定四分之一波长部分的电气长度和特性阻抗。 (b) 如果四分之一波截面是 D = 2.5 cm 的两线线路，并且导线嵌入 $\varepsilon_r = 2.6$ 的聚苯乙烯中，则确定四分之一波截面的物理长度和两根导线的半径。

2.41 长度为 l = 0.375λ 的 50 Ω 无损线路将具有 $\widetilde{V}_g = 300$ V 和 $Z_g = 50\ \Omega$ 的 300 MHz 发生器连接到负载 Z_L。确定通过负载的时域电流：

(a) $Z_L = (50 - j50)\ \Omega$

*(b) $Z_L = 50\ \Omega$

(c) $Z_L = 0$ (short circuit)

对于 (a) 部分，通过从模块 2.4 生成的输出结果中推导出所需的信息来验证您的结果。

第 2-9 节：无损线路上的功率流

2.42 $\widetilde{V}_g = 300$ V 和 $Z_g = 50\ \Omega$ 的发电机通过长度为 l = 0.15λ 的 50 Ω 无损线路连接到负载 $Z_L = 75\ \Omega$。

*(a) 计算 Z_{in}，即发电机端线路的输入阻抗。 (b) 计算 $\widetilde{V}_i$ 和 $\widetilde{V}_i$。 (c) 计算传输到线路的时间平均功率 $P_{in} = \frac{1}{2}\mathfrak{Re}[\widetilde{V}_i \widetilde{I}_i^*]$。 (d) 计算 $\widetilde{V}_L$、$\widetilde{I}_L$ 以及输送到负载的时间平均功率 $P_L = \frac{1}{2}\mathfrak{Re}[\widetilde{V}_L \widetilde{I}_L^*]$。 P_{in} 与 P_L 相比如何？解释。 (e) 计算发电机提供的时间平均功率 P_g 和 Z_g 中消耗的时间平均功率。能量守恒定律是否得到满足？

2.43 如果图 P2.43 所示的双天线配置连接到 $\widetilde{V}_g = 250$ V 和 $Z_g = 50\ \Omega$ 的发电机，则向每个天线传输的平均功率是多少？

*2.44 对于图 P2.44 所示的电路，计算平均入射功率、平均反射功率和传输到无限 100 Ω 线的平均功率。 λ /2 线是无损的，而无限长的线是轻微有损的。 （提示：只要 $\alpha \neq 0$，无限长线路的输入阻抗就等于其特性阻抗。）

2.45 图 P2.45 所示的电路由一条 100 Ω 无损传输线组成，该传输线终止于 $Z_L = (50 + j100)\ \Omega$ 的负载。如果测量负载电压的峰值为 $|\widetilde{V}_L| = 12$ V，则确定：

*(a) 负载中消耗的时间平均功率，(b) 线路上入射的时间平均功率，以及 (c) 负载反射的时间平均功率。